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Technical-economic and environmental assessment of marine biofuels produced in Brazil

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ABSTRACT

This study compared the technical-economic and life cycle environmental (LCA) performance of a bulk carrier, powered by different biofuels that could potentially be produced in Brazil, with a ship powered by low-sulfur fuel oil (LSFO). Six biofuel pathways and two power systems (two-stroke diesel engine and solid oxide fuel cell) were evaluated. The technical-economic assessment estimated the minimum fuel selling price (MFSP), and the impacts on the levelized cost of shipping (LCOS). All biofuels showed reductions ranging from 63.12% to 89.98% in GHG emissions per t.km compared to LSFO. Biofuels also performed better in terms of fine particulate matter formation and fossil resource scarcity, while results for the other environmental categories varied according to the production pathway. From an economic perspective, all biofuels showed higher MFSP (2.46–3.75 times) and LCOS (1.58–3.08 times) compared to LSFO. The results showed the potential of biofuels to reduce the carbon footprint and the economic challenges to overcome. In this sense, governmental programs and incentives can allow marine biofuels bridge the competitivity gap with fossil fuels. In the short-term, the adoption of different strategies, such as offsetting schemes, may be necessary for the international shipping decarbonization.

1. Introduction

International shipping is an important sector for the global economy, carrying around 80% (by volume) of international trade (Taljegard et al., 2014). At the same time, this sector is also responsible for significant impacts, contributing to about 2–3% of the world's greenhouse gas (GHG) emissions, 10–15% nitrogen oxides (NOx) and 4–9% of sulfur oxides (SOx) emissions (Hsieh and Felby, 2017). Seeking to mitigate the environmental impacts caused by the sector, the International Maritime Organization (IMO) reduced the allowable sulfur content in marine fuels from 3.5% to 0.5% in 2020 (IMO, 2021a) and has been imposing increasingly strict limits on NOx emissions (IMO, 2021b). In 2023, the organization adopted the 2023 IMO GHG Strategy, aiming, in particular, to reduce the carbon intensity of international shipping by at least 40% by 2030, compared to 2008, and reach net-zero emissions by or around 2050 (IMO et al., 2023). In this sense, biofuels may play an important role in reducing these impacts (IPCC, 2023; IMO, 2020), with Brazil – as one of the main global leaders in biofuel production – being among the most relevant potential suppliers.

The adoption of biofuels in the shipping sector is a topic of growing interest. Recent studies have addressed the technical and adaptability aspects of using different fuels in ships (Carvalho et al., 2021a; Kesieme et al., 2019; Wei, 2021) as well as the technical-economic and environmental performance of different marine biofuels (Carvalho et al., 2021b; Schaeffer et al., 2020; Tan et al., 2021; Tanzer et al., 2019; Li et al., 2022). The options are diverse (Fig. S1, Supporting Information) with different levels of maturity for ships application.

Given such wide range of options, further investigations are needed to expand knowledge on the technical-economic and environmental performance of different bio-based fuels, considering specific regional conditions and medium to long-term options, which are not extensively studied in the literature. Moreover, most studies consider fuel production costs, without extending the analysis to onboard technology, necessary for understanding potential impacts on freight costs. This is a significant indicator given the relevance of maritime transportation to the global economy and can support decision-making by considering a broader view of the economic impacts.

The present study analyzed some options considering biofuels with potential for short-term application as well as options with potential for medium and long-term applications. Among the different options, distillate biofuels like hydrotreated vegetable oil (HVO), hydrotreated pyrolysis oil (HDPO) and Fischer-Tropsch diesel (FT-diesel) have been described as being some of the most promising options, given their high

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Abbreviations

1G	first-generation
2G	second-generation
LSFO	low-sulfur fuel oil
HVO	hydrotreated vegetable oil
HDPO	hydrotreated pyrolysis oil
EtOH	ethanol
ICE	internal combustion engine
SOFC	solid oxide fuel cell
MFSP	minimum fuel selling price
LCOS	levelized cost of shipping
AC	abatement cost
SC	sugarcane straw
EC	eucalyptus forest residues
LUC	land use change

energy density and compatibility with existing infrastructures on ships (Carvalho et al., 2021a), and therefore considered as good options for short and medium term. In this study, HVO was chosen as a short-term biofuel candidate given its technology readiness level (TRL) of 9 (IPCC, 2023) and potential production in Brazil. HDPO in turn was chosen as a second-generation drop-in biofuel candidate, given that there are expectations that the fuel will have a better environmental performance, considering that can be produced from lignocellulosic residues. Therefore, HDPO can serve as a good reference for comparison with HVO, which despite its potential for short-term use, presents concerns regarding emissions from land use change. Furthermore, compared to FT-diesel, HDPO is expect to have a lower cost (IPCC, 2023; Carvalho et al., 2021a; Tanzer et al., 2019) and a fuel quality that can be more compatible with marine engines, which has fewer constraints in terms of fuel quality compared to other modes of transportation (IPCC, 2023).

For the medium to long term, however, preferences are expected to shift towards ammonia (NH_3) (DNVGL, 2021), which can account for around 45% of the global energy demand for the maritime sector in 2050 (IEA and Net, 2021). NH_3 can be consumed in 2-stroke diesel engines (with modifications) or in fuel cells, such as the solid oxide fuel cell (SOFC), which can reduce harmful emissions during ship operation. Flexibility, in terms of production pathways (from hydrogen via water electrolysis or biomass gasification), is an important advantage of ammonia, although the water electrolysis-based route has been more explored in the literature (Carvalho et al., 2021a; Kanchiralla et al., 2022; ALFA LAVAL et al., 2020; Hansson et al., 2020; Kim et al., 2020). In this sense, NH_3 from biomass-based route consumed in ICE and SOFC was chosen in this analysis as a medium to long-term biofuel option.

The use of alcohols as a marine fuel, especially methanol, is also an important option, with many pilot and demonstration projects being conducted on large ships (Global Maritime Forum, 2021). Ethanol, on the other hand, is less discussed in the literature due to its higher price in the international market compared to methanol, as well as the argued availability limitation of land, water and energy necessary for its production (Ellis and Tanneberger, 2018). Despite limitations for maritime applications, ethanol (EtOH) should be explored as a potential option, particularly in the Brazilian context, given the large production capacity that is already in place, and potential long-term reductions in demand, arising from light road vehicle fleet electrification. Furthermore, ethanol-fuel-cell-powered vessels are effective routes for reducing harmful emissions for ship operations (Carvalho et al., 2021a). In this sense, ethanol in SOFC was explored in this study as a long-term option.

This study assessed the environmental life cycle and technical-economic performance of cargo transports in bulk carriers, by consuming different fuels that could potentially be produced in Brazil. For ships equipped with a diesel two-stroke internal-combustion engine (ICE), the following pathways were analyzed: (1) Low Sulfur Fuel Oil (LSFO/ICE), (2) hydrotreated vegetable oil produced from soybean oil (S-HVO/ICE) and palm oil (P-HVO/ICE), (3) hydrotreated pyrolysis oil produced from the fast pyrolysis of sugarcane straw (SC-HDPO/ICE) and eucalyptus forest residues (EC-HDPO/ICE) and (4) ammonia produced from H_2 from sugarcane straw gasification (SC–NH_3/ICE) and eucalyptus forest residues gasification (EC-NH_3/ICE). The consumption of biofuels in SOFC were also evaluated, such as (5) ammonia (EC-NH_3/SOFC and SC-NH_3/SOFC) and (6) sugarcane ethanol (EtOH/SOFC) pathways.

The environmental analysis employed the LCA technique, and the technical-economic analysis evaluated the minimum fuel selling price (MFSP) of each pathway, and its impacts on the levelized cost of shipping (LCOS). Lastly, the Abatement Cost (AC) was calculated allowing to identify the hot spots and most promising alternatives, in terms of environmental impact mitigations associated with shipping, as well as the economic efforts needed to make them competitive with conventional fossil fuels.

This study is relevant in face of the urgency and the challenges associated with the international shipping decarbonization. The assembly of the life cycle inventories and the environmental-economic results can be used for comparative purposes with other (bio)fuel pathways to support the decision making process, as well as subsidize the development of regulations and public policies, such as IMO GHG Strategy and RenovaBio (BRASIL, 2017). According to the IMO, the use of alternative fuels with low carbon footprint is the main driver for reducing GHG emissions, potentially contributing to 64% of the reductions by 2050 (IMO, 2020). This scenario is further reinforced according to the IPCC Sixth Assessment Report, in which advanced biofuels has a mitigation potential of 73% in shipping (IPCC, 2023). Currently, Brazil is one of the world's leading producers of biofuels, with the potential to become a key supplier of sustainable fuels for the international shipping. Therefore, understanding the economic and environmental performance of some options potentially produced in the country becomes relevant for the international scenario.

2. Material and methods

The methodology framework used in this study is shown in Fig. 1. Using the life cycle inventory (LCI) it was possible to perform the Life Cycle Impact Assessment (LCIA), as well as to build the cost frameworks for biofuel production and for ship operations. These cost frameworks allowed to estimate the MFSP in the port and the LCOS of each pathway. Finally, based on the LCOS and on the life cycle GHG emissions, the AC were calculated for each alternative relative to the LSFO/ICE (reference case), thus giving the incremental cost to be paid to mitigate 1 tCO_2eq.

2.1. Life cycle assessment

The LCA was used to compare the potential environmental impacts of the studied alternatives, helping understand the role of biofuels in shipping decarbonization, as well as to identify their contributions by considering other environmental impacts. The study considered an attributional and well-to-propeller approach. The LCIA was modelled following the Recipe 2016 Midpoint (H) method for fifteen environmental impact categories, which are: global warming, fine particulate matter formation, fossil resource scarcity, mineral resource scarcity, ozone formation (human health), ozone formation (terrestrial ecosystems), human toxicity (carcinogenic), human toxicity (non-carcinogenic), freshwater ecotoxicity, marine ecotoxicity, freshwater eutrophication, marine eutrophication, terrestrial acidification, terrestrial ecotoxicity and stratospheric ozone depletion. This LCA study considered an attributional approach and defined the functional unit as 1 t km. The datasets were developed considering the Brazilian scenario. Therefore, life cycle inventories that assessed conditions in Brazil were preferably used as references. Additionally, some inventories referenced in international studies were adapted to consider Brazilian conditions,

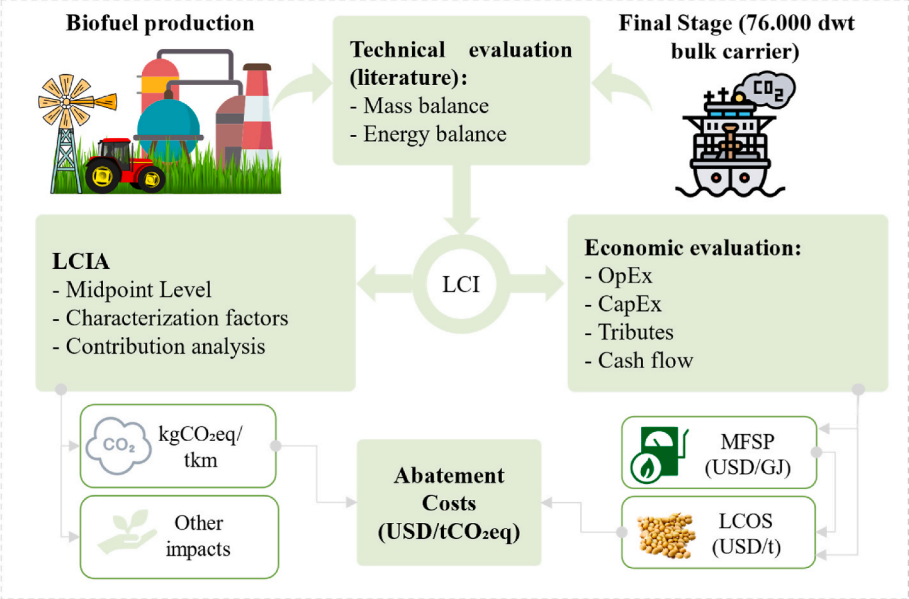


Fig. 1. Methodological framework employed in the study.

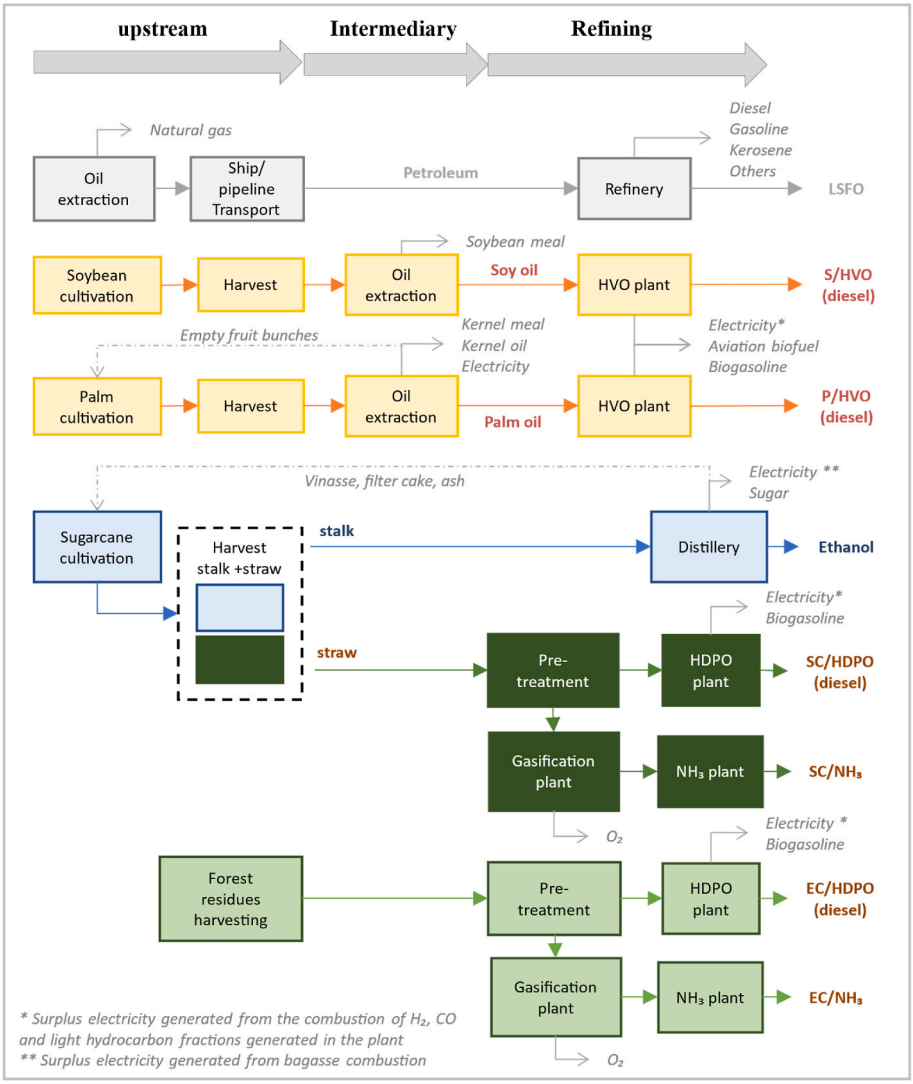


Fig. 2. Pathways for fuel production.

such as the electricity grid (see Table S52), biodiesel content in diesel (see Table S51), and logistic distances (further details in the following sections). All datasets can be consulted in more detail in the Supporting Information (SI).

Fig. 2 illustrates the studied pathways, as well as the main co-products and residues. Six feedstocks were considered (petroleum, soy, palm, sugarcane, sugarcane straw and eucalyptus forest residues) producing 5 marine fuels (LSFO, HVO, EtOH, HDPO (diesel fraction) and NH_3). In this study S-HVO, P-HVO and EtOH were classified as first-generation biofuels (1G) since they were produced from food crops. HDPO and NH_3 were classified as second-generation (2G) biofuels, as they were produced from lignocellulosic waste. The different lifecycle stages are explained in more detail in the following sections.

2.1.1. Upstream and intermediary stages

Soybean production was modelled as a monoculture in *Mato Grosso* state (Capaz et al., 2021a; IBICT/SICV, 2020; Matsuura and Picoli, 2018), taking an average distance between the soy oil extraction plant and the farm as 400 km (one way), which was driven by trucks (Capaz et al., 2021a) (see Table S15). It was considered, in the intermediate stage, an oil extraction plant via hexane co-producing soybean oil and meal (Capaz et al., 2021a) (Table S16).

Palm cultivation was modelled as a monoculture system in *Pará* state (Capaz et al., 2021a; Agropalma, 2019). The distance between the farm to the milling plant was 32 km (one way), which was also driven by truck (Capaz et al., 2021a). The milling plant produces palm oil, cake, palm kernel oil, and surplus electricity (Capaz et al., 2021a). The surplus electricity is generated in cogeneration boilers via the combustion of palm fibers and shells (Capaz et al., 2021a; Agropalma, 2019). It was also considered electricity generation from biogas burning, produced via the anaerobic digestion of palm oil mill effluent (POME) in a closed pond system (Capaz et al., 2021a; Agropalma, 2019). More details in Tables S20 and S21 in the SI.

Sugarcane, in turn, was modelled as a monoculture system in *São Paulo* state (Capaz et al., 2021a; Carvalho et al., 2021d; Cavalett et al., 2016). Stalk and straw were taken as being harvested together (integral harvest) (SUCRE, 2020), with a distance between the farm and the plant of 21 km (Seabra et al., 2011) (see Table S41).

For eucalyptus forest residues, the inventory included only operations after log harvesting (Nogueira et al., 2021), considering a distance of 80 km between the farm and the waste processing unit (de Almeida, 2016). It is worth mentioning that this study treated both eucalyptus forest residues and sugarcane straw as being residues. This implies that the environmental burdens were allocated only to the main products, i.e., eucalyptus logs and sugarcane stalks, respectively. The impacts associated with the residues, in turn, came only from collection and transportation. More details in Tables S25 and S29 in the SI.

Finally, for the LSFO, this study took background data on the extraction of oil and natural gas from offshore platforms (Ecoinvent, 2017). The oil transportation between the offshore platforms and the port was modelled using an oil tanker, with an average distance of 310 km (one way). The oil transportation from the ports to the refinery was modelled using pipelines, with a total distance of 236 km (between the *São Sebastião-Guararema-REPLAN* terminals) (Transpetro, 2021).

2.1.2. Refining stage

The same industrial processing route to produce HVO was considered for both oilseed pathways, but with different values for H_2 consumption in the hydrogenation stage. It was considered an external H_2 production via natural gas steam reforming (Capaz et al., 2021a). For soybeans, the distance between the oil extraction plant to the HVO plant was 1200 km (one way), which corresponds to the approximate distance between the municipality of *Rondonópolis-MT* (one of the largest producers of soybean oil in Brazil) (ANP, 2022) and the REPLAN refinery. More details in Table S17 in the SI. For palm oil, the distance was 4350 km (one way), which was covered by ships, corresponding to the distance between the ports of *Belém-PA* and *Santos-SP* (Sea-Distances, n.d.). More details in Table S22 in the SI. The distance between the HVO plant and the port was 220 km (one way), which was driven by trucks, which is the approximate distance from REPLAN to the *Santos* Port.

For ethanol, a typical Brazilian plant with the co-production of ethanol, electricity, and sugar was considered (Seabra et al., 2011) (Table S42). The HDPO plant, in turn, was modelled considering a fast pyrolysis plant in combination with hydrotreatment and hydrocracking systems producing fractions of diesel and gasoline (Dutta et al., 2015a, 2018a), with internal H_2 production (see Tables S26 and S30). The NH_3 plant was modelled using the Haber-Bosch process, consuming H_2 from biomass gasification (Spath et al., 2005). More details in Tables S33 and S37 in the SI. In both cases it was assumed that the 2G units were adjacent to the main units (ethanol plant and pulp and paper plant). The distance between the ethanol, HDPO, and NH_3 plants to the *Santos* Port was taken as being 500 km (one way), travelled entirely by truck. Table 1 presents a summary of the life cycle inventory for the different pathways (More details in the SI).

2.1.3. Use stage

This study made a distinction between the ship's operation emissions close to the coast and in the open sea. For ships equipped with ICE, the emissions profiles were based on literature data for burning LSFO (IMO, 2020; EEA, 2019), adjusting emissions factors for biofuels (further details in the SI).

2.1.4. Capital goods

For capital goods, the impacts of the manufacturing stage for ships, the power system, the auxiliary systems, and the industrial units were considered. In all cases, secondary databases were used without adaptations (Ecoinvent, 2017).

2.1.5. Direct land use change (dLUC)

dLUC emission scenarios were explored for 1G biofuels. In this case, variations in carbon stocks were considered for 4 different land use categories: annual crops, sugarcane crops, perennial crops, and pasture. For soybeans, the carbon stock of annual crops was assumed and for palm, the stock of perennial crops was considered. Carbon stocks for each land use category were obtained from the BRLUC v.1.3 tool (Novaes et al., 2017) for different Brazilian states, i.e., *São Paulo*, *Mato Grosso*, and *Pará* (Table 2). It was considered the sugarcane expansion in *São Paulo* state (SP), palm in *Pará* state (PA), and soybeans in *Mato Grosso* state (MT). Indirect land use change was not considered in this study.

2.2. Technical-economic assessment

As outlined in Fig. 3, the technical-economic assessment was divided into two stages: (i) calculation of the MFSP and (ii) calculation of the LCOS. The MFSP and LCOS were calculated as the minimum selling price for fuel and freight services, respectively, needed to return net present value (NPV) equal to zero in the projected cash flows. The calculation was performed considering a fixed discount rate, capital (CapEx) and operational expenditures (OpEx), fees, taxes, as well as revenues from the sale of co-products (if any), over the entire project lifetime. The 2019 U.S. Dollar currency was used in the calculations (USD_{2019}).

2.2.1. Minimum fuel selling price (MFSP)

The MFSP calculation was performed only for HDPO, NH_3, and HVO. For EtOH, a price of 0.45 USD/L was taken as an average value, considering the period from 2015 to August 2021 in the Brazilian market (CEPEA/USP, 2021). For LSFO, the average value of 500.00 USD/t was taken, sold in the port of *Santos/BR* in 2021 (Ship and Bunker, 2021). In both cases, the amounts were obtained from net calculations after Brazilian taxes, e.g., PIS/COFINS.

The first step in the MFSP calculation was to estimate the capital

Table 1
Main processes and references for the life cycle inventory of pathways.

Upstream Stage								
Pathways	S-HVO	P-HVO	EtOH	SC-HDPO	SC-NH3	EC-HDPO	EC-NH3	Unit
Main references	(Capaz et al., 2021a; IBICT/SICV, 2020; Matsuura and Picoli, 2018)	(Capaz et al., 2021a; Agropalma, 2019)	(Capaz et al., 2021a; Carvalho et al., 2021d; Cavalett et al., 2016)	SUCRE, 2020	SUCRE, 2020	Nogueira et al. (2021)	Nogueira et al. (2021)	
Feedstock	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	t
Inputs								
diesel, burned in agricultural machinery	5.35E+00	5.24E-02	2.04E+00	1.71E-01	1.71E-01	5.34E+00	5.34E+00	kg
nitrogen fertiliser, as N	2.24E+00	7.30E+00	1.07E+00	–	–	–	–	kg
organic fertiliser, as N	–	4.30E-01	2.84E-01	–	–	–	–	kg
phosphate fertiliser, as P2O5	2.24E+01	1.80E+00	4.06E-01	–	–	–	–	kg
potassium chloride, as K2O	2.02E+01	1.30E+01	9.75E-01	–	–	–	–	kg
other products	5.38E+01	7.43E-01	7.55E+00	–	–	–	–	kg
transport, freight, lorry >32 metric ton, EURO4	4.00E+02	3.20E+01	(a)	(a)	(a)	8.00E+01	8.00E+01	t*km
crop residues	1.61E+03	8.95E-03 (as N)	1.65E+02	–	–	–	–	t
Emissions	air, soil, water	air, soil, water	air, soil, water	air	air	air	air	
Intermediate Stage								
Main references	Capaz et al. (2021a)	Capaz et al. (2021a)	–	–	–	–	–	
SVO	7.17E+03	6.61E+03	–	–	–	–	–	MJ
coproducts	1.07E+04	1.07E+03	–	–	–	–	–	MJ
surplus electricity	–	7.54E+00	–	–	–	–	–	kWh
Inputs								
feedstock	1.00E+00	1.00E+00	–	–	–	–	–	
heat	2.88E+02	1.72E+03	–	–	–	–	–	MJ
electricity {BR}	4.83E+01	–	–	–	–	–	–	kWh
electricity {POME biogas}	–	8.04E+01	–	–	–	–	–	kWh
transport, freight, lorry >32 metric ton, EURO4	1.20E+03	–	–	–	–	–	–	
transport, freight, sea, transoceanic ship	–	4.35E+03	–	–	–	–	–	
other products	3.01E+02	–	–	–	–	–	–	kg
Refining Stage								
Main references	Pearlson (2011)	Pearlson (2011)	Seabra et al. (2011)	(Dutta et al., 2015a, 2018a)	Spath et al. (2005)	(Dutta et al., 2015a, 2018a)	Spath et al. (2005)	
Marine biofuel	2.96E+04	2.96E+04	8.92E+02	4.89E+03	6.59E+03	5.48E+03	7.35E+03	MJ
coproducts	6.44E+03	6.44E+03	9.14E+02	4.46E+03	–	4.96E+03	–	MJ
surplus electricity	3.52E+02	3.85E+02	3.23E+01	3.29E+01	–	3.68E+01	–	kWh
Inputs								
feedstock (SVO or biomass)	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	t
heat {offgas combustion}	2.68E+03	2.68E+03	–	7.41E+02	1.09E+03	7.41E+02	1.09E+03	MJ
biomass combustion	–	–	1.30E+02	1.20E+02	2.20E+02	1.20E+02	2.20E+02	kg
electricity {BR}	–	–	–	–	1.39E+02	–	1.55E+02	kWh
hydrogen {from natural gas}	2.70E-02	2.27E-02	–	–	–	–	–	t
transport, freight, lorry >32 metric ton, EURO4	1.50E+02	1.50E+02	1.62E+01	6.99E+01	1.77E+02	7.83E+01	1.98E+02	t*km
Other products	9.00E+02	9.00E+02	1.35E+03	3.85E+02	2.43E+02	3.85E+02	2.43E+02	kg

(a) Biomass transport already considered in diesel consumption.

costs for the producing plants. The OpEx, in turn, were divided into fixed costs (fixed O&M), which do not depend on the amount of biofuel produced (e.g., maintenance, insurance, and labor costs), and variable costs (variable O&M), which depend on the amount of biofuel produced in the year (e.g. feedstock consumption, electricity, hydrogen, catalysts, etc.). General economic parameters are expressed in Table 3 and the calculated capital and operational costs are expressed in Table 4. These data were obtained from literature (see more details in Tables S4, S6 and S7 in the SI).

With the biorefinery CapEx and OpEx, it was possible to calculate the MFSP, according to the characteristics of the enterprise, the financing scheme, and taxes. In cases where electricity was produced as a co-

Table 2
Carbon stocks (below and above ground) for each crop and region (Novaes et al., 2017).

Agricultural culture	Unit	SP	MT	PA
annual crops	tC/ha	36.19	39.08	42.09
Sugarcane crops	tC/ha	44.59	47.18	50.43
perennial crops	tC/ha	74.57	82.03	102.16
pasture	tC/ha	47.00	49.62	52.82

product, it was considered an average sales price of 0.25 BRL/kWh, obtained from the energy tariff in Brazilian markets for the 2020–2021 period, without grid access charges and taxes (PIS/COFINS and ICMS) (ANEEL, 2021). The value was converted into dollars, considering the 2019 exchange rate at 3.95 BRL/USD.

2.2.2. Levelized cost of shipping (LCOS)

The LCOS was calculated considering a 76,000 dwt bulk carrier shipping soybean (770.00 kg/m^3). The estimates accounted for the capital costs of the ship, as well as the operational and voyage costs. The ship's CapEx estimates were divided into two components: (i) ship costs without a power system and storage tanks (a common value for all studied alternatives) and (ii) power systems and storage tank costs for each technology/fuel. The capital costs for the equipment were obtained from literature, and the values were adjusted to the 2019 dollar rate, using the CEPCI index.

Voyage costs accounted for fuel costs and port fees. Fuel costs were based on MFSP calculations and the ship's annual fuel consumption. To calculate fuel consumption, the ship's energy demands were considered under average operating conditions (see Table S3 in the SI). Operating costs considered costs with crew salaries, consumables, maintenance, insurance, and general costs, and were all based on data from literature (Stopford, 2009a).

The loss of cargo capacity on alternative ships powered by biofuels was also considered in the LCOS calculation. The study considered loss of space due to biofuel storage tank and power system installations on the ships relative to the reference case (LSFO/ICE). The difference in occupied volume/mass was then deducted from the ship's transport capacity (further information can be found in the SI). Table 5 presents the interest rate, CapEx and OpEx calculated values of the ships (more details see Tables S10 and S11 in the SI).

2.2.3. Abatement cost (AC)

It was possible to calculate the GHG emissions abatement costs for each alternative relative to the reference case using the LCOS and life-cycle GHG emission calculations, to give the additional amount to be paid for 1 ton of mitigated CO_2eq. The AC calculation for each pathway (AC_i [USD/tCO_2eq]) is expressed using Equation (1), as a function of the LCOS of the technology of interest ($LCOS_i$), its lifecycle GHG emissions (LCA_i), and the LCOS and lifecycle GHG emissions of the reference technology ($LCOS_{LSFO}$ and LCA_{LSFO}, respectively).

$$AC_i = \frac{LCOS_i - LCOS_{LSFO}}{LCA_{LSFO} - LCA_i} \quad \text{Equation 1}$$

2.3. Sensitivity and uncertainty analysis

The uncertainties surrounding the economic and environmental performance estimations were addressed using the Monte Carlo method, with OpenLCA (openLCA, 2021) (for LCA) and Crystal Ball (Oracle CrystalBall, n.d.) (for economic analyses) used as simulation tools. The method was applied to compare the LCIA results, and the MFSP, LCOS, and AC results. All data analyses and graphic compilations were performed in Excel (Microsoft Excel, 2021). Sensitivity analysis was also performed. In this case, the modeling defined the minimum and maximum values from the 1st and 99th percentiles, respectively, derived from the probability distributions of each parameter used in the Monte Carlo method. Sensitivity results can be consulted in the supporting information.

3. Results and discussion

3.1. Life Cycle Impact Assessment

Fig. 4 illustrates the LCIA results for the Global Warming category without considering dLUC emissions. It is possible to notice a reduction

Table 3
Main economic variables for nth plant[1].

Parameters	Unit	Value
Economic parameters		
Exchange rate (reference year 2019)	BRL/USD	3.95
Depreciation time (linear)	years	10.00
Interest rate (1G biofuel plant)	–	13.10%
Interest rate (2G biofuel plant)	–	16.10%
BNDES Financing Conditions[2]		
1G plant loan interest rate	–	11.00%
2G plant loan interest rate	–	13.15%
% Project financed	–	80.00%
Grace period after start of operation	–	Up to 6 months
Payment term	Years	12.00
Charges on Real Profit[3]		
CSLL and Income tax	–	34.00%
Plant construction time[4]		
Year −2	%TCI	30.00%
Year −1	%TCI	50.00%
Year 0	%TCI	20.00%

[1] More details see Table S4 in SI.
[2] **Financing conditions** based on BNDES (The Brazilian Development Bank) (BNDES, 2021).
[3] **CSLL**: Brazilian tax intended to finance Social Security. **Income tax**: a single rate of 25.00% was assumed.
[4] Plant construction time based on (de Jong et al., 2015a). **TCI**: Total Capital Investment.

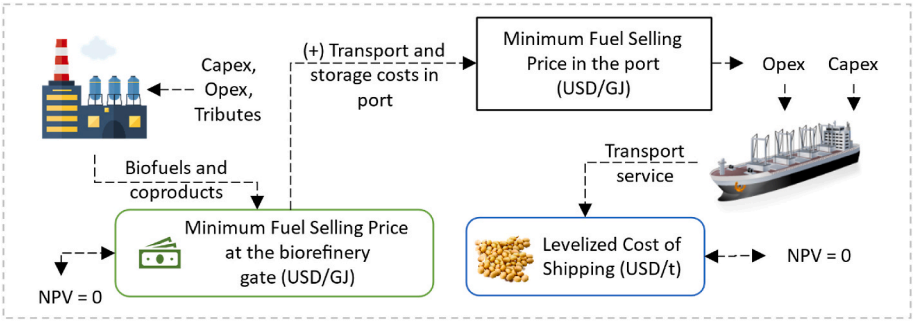


Fig. 3. Technical-economic assessment framework.

Table 4
Capital costs and Operational costs for different biofuels pathways.[1].

Parameters	Unit	S-HVO	P-HVO	EC-HDPO	SC-HDPO	EC-NH_3	SC-NH_3
Production scale[2]	t feedstock/year	128,700	128,700	173,478	240,000	173,478	240,000
Plant lifetime	Years	25.00	25.00	25.00	25.00	25.00	25.00
Direct Costs (%TPEC)[3]	–	147.00%	147.00%	147.00%	147.00%	147.00%	147.00%
Indirect Costs (% TPEC)[4]	–	126.00%	126.00%	126.00%	126.00%	126.00%	126.00%
Startup costs (%TPC)[5]	–	6.00%	6.00%	6.00%	6.00%	6.00%	6.00%
Brazilian Location Factor (%TPC)[6]	–	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%
Land cost (% TPEC)[7]	–	6.00%	6.00%	6.00%	6.00%	6.00%	6.00%
Working capital (%FCI)[8]	-	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
CapEx (TCI)	**MM USD_{2019}**	**241.64**	**241.64**	**347.01**	**454.41**	**248.04**	**300.72**
Employee wages and benefits[9]	MM USD_{2019}	1.89	1.89	0.66	0.95	0.66	0.95
Maintenance – (% TCI)[10]	–	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%
Fees and insurance – (% TCI)[11]	–	0.70%	0.70%	0.70%	0.70%	0.70%	0.70%
Annual fixed O&M costs	**MM USD_{2019}**	**10.84**	**10.84**	**13.50**	**17.76**	**9.83**	**12.07**
Vegetable oil[12]	USD/t	852.41	763.58	–	–	–	–
Residual biomass[13]	USD/tbs	–	–	24.95	20.25	24.95	20.25
Variable annual O&M[14]	**MM USD_{2019}**	**115.52**	**103.31**	**7.12**	**8.83**	**9.53**	**11.52**
Fuel storage + transport cost[15]	**USD/GJ**	**0.56**	**0.56**	**0.77**	**0.77**	**2.37**	**2.37**

[1] More details see Tables S6 and S7 in SI.
[2] **Soybean and palm oil** (Cervi et al., 2020); **Eucalyptus forest residues**: value in dry basis adjusted by the annual plant capacity factor, considering the production of residues in an eucalyptus cultivation area of 31,482 ha (Nogueira et al., 2021), a residue production factor of 8.20 t/ha (10.4% moisture) (Pincelli et al., 2017) and a collection factor of 75.00% (Nogueira et al., 2021); **Sugarcane straw**: value in dry basis adjusted by the annual plant capacity factor, considering an ethanol plant that processes 4,000,000 tons of sugarcane per year, a straw production factor of 120.00 t(db)/t of sugarcane and an assumed collection factor of 50.00%.
[3] Based on (Spath et al., 2005; Jones et al., 2009; Villanueva Perales et al., 2011).
[4] Based on (Spath et al., 2005; Jones et al., 2009; Dutta et al., 2018b).
[5] Based on (de Jong et al., 2015b).
[6] Based on (Tanzer et al., 2019).
[7] not depreciable (Spath et al., 2005).
[8] Based on (Dutta et al., 2018b).
[9] More details see Table S7 in SI.
[10] Based on (Dutta et al., 2018b).
[11] Considering contributions to social security, FGTS, severance pay and employee benefits (in BRL) (IBGE, 2020). Amount converted to dollar considering the exchange rate of 3.95BRL/USD.
[12] **Vegetable oils**: considered average international price in the period from Jul/2011 to Jun/2021 (IndexMundi, 2021).
[13] **Eucalyptus forest residue**: cost of 0.99 USD/GJ (Carvalho et al., 2021b); **Sugarcane straw**: cost of 80.00 BRL/t(db) for integral harvesting (1USD = 3.95BRL).
[14] All costs were based on the literature. For costs in BRL, amounts referring to years prior to 2019 were corrected by the IGP-DI index (FGV and FGVDados, 2021). For values in USD, costs were adjusted by the Consumer Price Index (CPI) (World Bank, 2021). In addition to biomass costs, the following costs were considered. **Electricity**: average cost for center-south region in Brazil (0.60BRL/kWh) (ANEEL, 2021); **Water**: considered an average price of 3.06 USD/m^3 (IBGE, 2018); **Hydrogen**: considered a cost of 1395 USD/tH2 for hydrogen produced from the steam reforming of natural gas (Capaz et al., 2021b); **Waste treatment**: 33.00USD/t (USD1998) for solid waste and 0.02USD/gal (USD2011) for wastewater (Dutta et al., 2018b); **Chemicals** (Spath et al., 2005); **Solvents (selexol)** (Im et al., 2015); **Fluidized bed material** (Dutta et al., 2018b); **catalysts** (Spath et al., 2005; Dutta et al., 2015b, 2018b; Zhang et al., 2020).
[15] More details see Table S8 in SI.

Table 5
CapEx and OpEx estimation for ships equipped with ICE.[1].

Parameters	Unit	LSFO	S-HVO	P-HVO	EC-HDPO	SC-HDPO	EC-NH3	SC-NH3	EC-NH3	SC-NH3	EtOH
		ICE							SOFC		
Interest rate[2]	–	8.60%	8.60%	8.60%	11.60%	11.60%	11.60%	11.60%	11.60%	11.60%	11.60%
CapEx[3]	MM USD_{2019}	30.55	30.46	30.46	30.27	30.27	41.82	41.82	58.80	62.57	62.57
OpEx[4]	MM USD_{2019}	1.11	1.09	1.09	1.09	1.09	1.20	1.20	1.29	1.32	1.32

[1] More details see Tables S10 and S11 in the SI.
[2] More details see Table S5 in SI.
[3] Considered a conventional ship cost of 30.00 MMUSD (Kretschmann et al., 2017), already assumed that direct and indirect costs are considered. Incremental costs were considered for the alternative ships powered by biofuels. See more details in Tables S9, S10 and S11 in SI.
[4] Operating costs were divided into 42.00% related to crew cost, 14.00% consumables, 16.00% maintenance, 12.00% insurance, 16.00% general costs (Stopford, 2009b).

in the biofuel pathway impacts compared to the reference case (LSFO/ICE, 6.26E-03 $kgCO_2eq/t.km$), with reductions ranging from 63.12% (S-HVO/ICE and P-HVO/ICE, approximately 2.31E-03 $kgCO_2eq/t.km$) to 89.98% (SC-HDPO/ICE, 6.28E-04 $kgCO_2eq/t.km$).

An important contribution of the ship's operation phase was observed in the LSFO/ICE pathway, associated with fossil CO_2 emissions into the atmosphere from LFSO combustion. For the 1G biofuel pathways (P-HVO/ICE, S-HVO/ICE and EtOH/SOFC), the biomass production stage had a relevant contribution, mainly associated with N_2O emissions from organic and mineral nitrogen fertilizers and fossil CO_2 emissions from soil amendment decomposition, e.g., urea and limestone, especially for sugarcane and soy cultivation. Another relevant impact came from diesel combustion (10.00% biodiesel – B10) in agricultural machinery used in biomass production and harvesting. For 2G biofuels, the impacts in the agricultural phase mainly came from the consumption of diesel (B10) in residue collections. Sugarcane straw

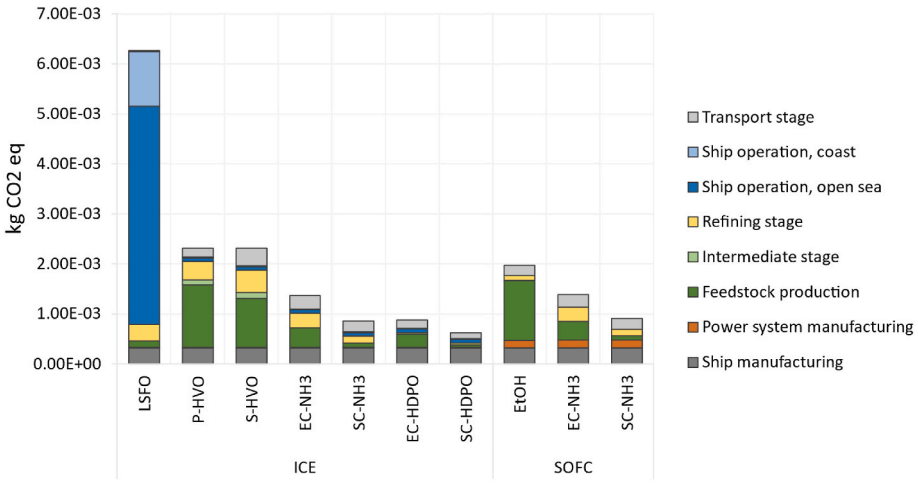


Fig. 4. LCIA (FU:1 t km) for the Global Warming category.

showed lower diesel consumption per ton of harvested biomass, compared to eucalyptus forest residues, and consequently, lower impacts in terms of GHG emissions in the agricultural stage.

It was possible to observe the relevant impacts of the refining stage (biofuel production stage) for the HVO and NH_3 pathways. For HVO, the impacts came mainly from CO_2 emissions from hydrogen production from external natural gas steam reforming. For NH_3, refining stage emissions came mainly from electricity consumption to feed the ammonia synthesis process, even though the low emissions factor of the Brazilian grid helped to attenuate GHG emissions in this stage.

For all biofuels, it was also identified significant impacts to GHG emissions from the transport stage. These impacts came mainly from fossil CO_2 emissions from diesel (B10) consumption in vehicles, especially trucks. For pathways using NH_3, the lower LHV of ammonia, and its lower density compared to other studied fuels combined for a more relevant impact from the transport phase in the lifecycle impacts. For the S-HVO/ICE pathway, in turn, greater distances in the Brazilian soybean value chain contributed to significant impacts to the transportation phase. These impacts can be mitigated using other less carbon-intensive means of transportation, e.g., via waterway, rail, or pipeline.

Finally, the ship manufacturing stage showed similar GHG emissions for all pathways, mainly associated with fossil CO_2 emissions in steel production. For the pathways equipped with SOFC, an additional GHG emission was identified in the fuel cell manufacturing stage, associated with electricity consumption and lanthanum oxide production. The NH_3/SOFC pathways did not show reductions in GHG emissions compared to ships equipped with ICEs powered by NH_3.

The results of the Monte Carlo simulation for Global Warming (without dLUC) are shown in Fig. S3 in the SI, where the box plots summarize the probabilistic distributions of the LCIA result, highlighting the median, percentiles (within a 95.00% confidence interval), and the base case values (i.e., the results obtained in the deterministic analysis). From the results obtained, all biofuel pathways presented lower lifecycle GHG emissions than the uncertainty range of the reference case, considering a 95.00% confidence interval. 2G biofuel pathways, in turn, presented quartile ranges below the quartile range of 1G biofuels, which reinforces the perception that 2G biofuels have better environmental performance.

Fig. S4 in the SI, in turn, shows the LCIA results for the Global Warming category considering different dLUC scenarios for 1G biofuels and comparing these different scenarios with the reference case (6.26E-03 $kgCO_2eq/t.km$). Depending on the dLUC scenario, it was possible to observe that the environmental benefits, in terms of GHG emissions, are significantly reduced or even nullified, e.g., soybean expansion over perennial crop areas (1.14E-02 $kgCO_2eq/t.km$). Opportunities for CO_2 removal were also observed, as exemplified by palm plantation expansion into pastureland (−2.45E-03 $kgCO_2eq/t.km$) and annual crop areas (−3.50E-03 $kgCO_2eq/t.km$), increasing carbon stocks with crop growth over time. These aspects underscore the relevance of sustainability and regulatory schemes to pay close attention to the direct land use changes associated with maritime biofuels, since these can constitute either an important source or sink of carbon. Furthermore, despite not being within the scope of this study, induced or indirect land use changes (iLUC) may also be triggered by the expansion of first-generation maritime biofuels, therefore requiring the adoption of appropriate mechanisms to limit the negative impacts from this market-driven effect.

All biofuel pathways feature lower impacts compared to the reference case in the categories for Fine particulate matter formation (2.69E-05 kgPM2.5eq/t.km) and Fossil resource scarcity (2.07E-03 kg oil eq/t.km) (see Fig. 5). Regardless of the fuel, the ICE based pathways always show the worst performance when it comes to Fine particulate matter formation, Ozone formation impacts on human health and on terrestrial ecosystems (Fig. S6), and Terrestrial acidification, mainly due to NO_x emissions from burning fuel in marine engines. In the NH_3/ICE pathway, large NO_x emission potential was identified, causing major impacts to the Ozone Formation category. For ships using SOFC, since it is a combustion-free process in operation, important reductions in lifecycle environmental impacts were identified compared to the reference case for categories like Fine particulate matter formation (67.95%–83.05%), Ozone formation, human health (89.99%–94.18%) and Terrestrial acidification (69.87%–85.25%), highlighting the potential of SOFC for reducing local emissions in ship operations.

For Freshwater ecotoxicity (Fig. 5), Human Carcinogenic Toxicity and Marine ecotoxicity (Fig. S5) it was possible to identify a significant impact coming from the ship manufacturing stage, mainly associated with emissions of metals into waterways in the steel production value chain. This stage was also significant for the Mineral resource scarcity (Fig. S5), with additional impacts for 1G biofuels in the feedstock production stage, mainly related to the consumption of phosphate fertilizers in the cultivation of biomass (produced from apatite). However, it is worth mentioning that the lifecycle inventory for the manufacturing stages relied on background datasets from Ecoinvent, which may not precisely portray the projected manufacturing conditions of the alternatives analyzed here.

For 1G biofuels, emissions of pesticides used in biomass cultivation into waterways resulted in greater impacts for Marine and Freshwater ecotoxicity. Other relevant burdens associated with biomass cultivation are N_2O emissions from the decomposition of organic and mineral nitrogen fertilizers and crop residues, which explain the higher scores obtained for Stratospheric Ozone Depletion and phosphorus and nitrate

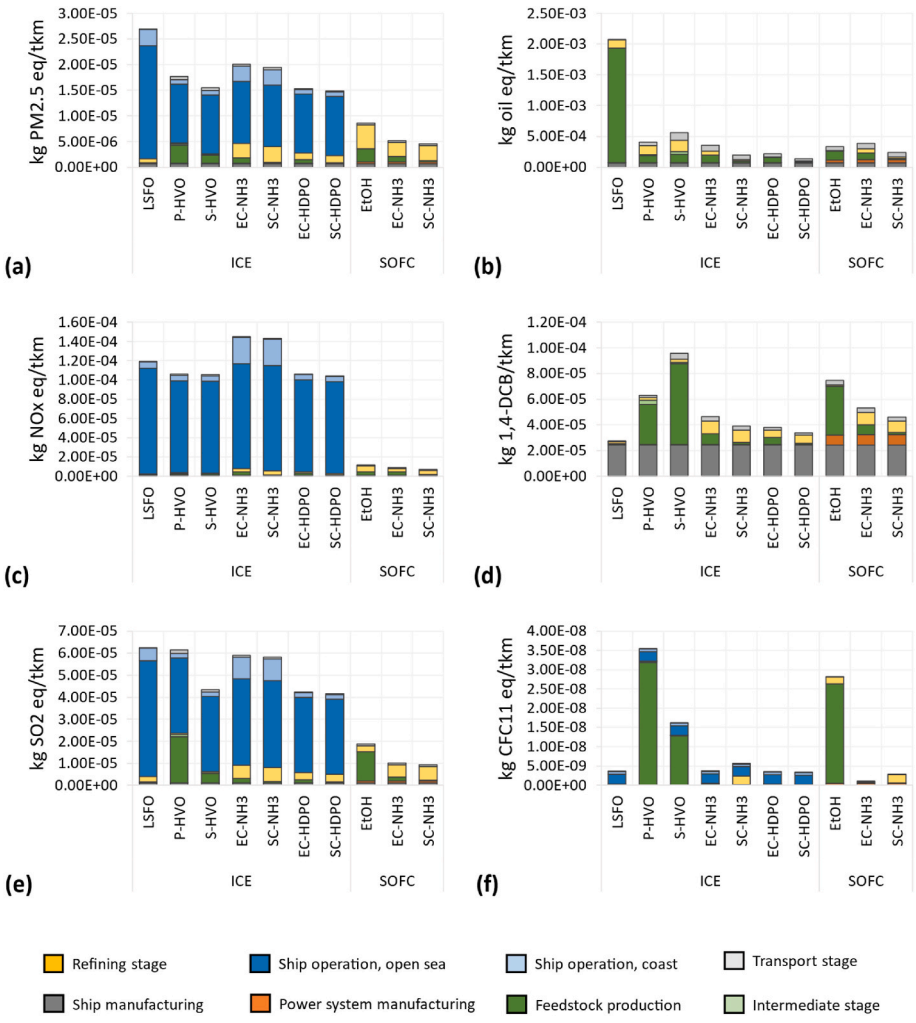


Fig. 5. LCIA (FU:1 t km) (a) Fine particulate matter formation; (b) Fossil resource scarcity; (c) Ozone formation, human health; (d) Freshwater ecotoxicity; (e) Terrestrial acidification; (f) Stratospheric ozone depletion.

emissions into water bodies, which contributed respectively to freshwater and marine eutrophication. Quantitative results can be consulted in Tables S60–S69 in the SI for each impact category and pathway.

3.2. Economic results

Regarding the economic performance, Fig. S7 in the SI shows that all biofuels presented MFSP values above the LSFO market price levels, with an important contribution from feedstock costs for HVO (vegetable oils), and from plant Capex and O&M costs for the 2G biofuels cases. Fig. 6 reinforces the economic challenges of biofuels by comparing the uncertainty results for MFSP at the port with the ranges for ethanol (CEPEA/USP, 2021) and LSFO (Ship and Bunker, 2021) market prices found in literature. In all scenarios, except for ethanol, the MFSP range for biofuel pathways was greater than the LSFO market price range. For HVO, the main uncertainties were due to changes in vegetable oil market prices, which is a typical (and critical) risk for biofuels that rely on more value-added feedstocks. For HDPO and ammonia, large uncertainties exist around the plant's capital costs.

Consequently, biofuel use leads to higher freight costs, as shown in Fig. 7. Furthermore, capital and replacement costs for SOFC also play relevant roles in the LCOS results, which pose significant economic drawbacks compared to ICE-based ships. This corroborates the

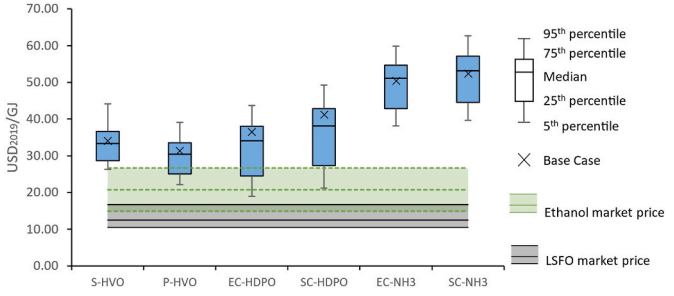


Fig. 6. Monte Carlo results for MFSP of the analyzed fuels in comparison with ethanol and LSFO price ranges.

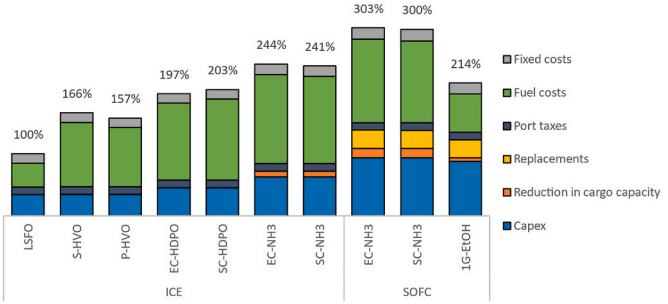


Fig. 7. Levelized cost of shipping (LCOS) relative to the reference case (LSFO).

understanding that fuel cells for maritime applications may become economically viable only over the long term, particularly for ships that have more intense activities in coastal areas, where the local pollutant emissions are a relevant problem. For ammonia, an additional issue was the reduction of the cargo capacity due to the storage tank installations. The Monte Carlo simulation results for the LCOS in each pathway, considering soybean transportation in a 76.000 dwt bulk carrier can be consulted in Fig. S8 in the SI.

3.3. Abatement costs

The relationship between GHG lifecycle emissions and LCOS indicators for each pathway was identified by calculating the AC, as shown in Fig. 8, thereby showing the amount to be paid for 1 tCO_2eq mitigations relative to the reference case (LSFO/ICE).

The lowest ACs levels were identified for drop-in biofuels in ICEs, like S-HVO/ICE (421.64 USD_{2019}/tCO_2eq), P-HVO/ICE (367.61 USD_{2019}/tCO_2eq), EC-HDPO/ICE (452.71 USD_{2019}/tCO_2eq) and SC-HDPO/ICE (462.04 USD_{2019}/tCO_2eq). HDPO pathways showed potential above 85.00% for reducing GHG emissions relative to the reference case. For NH_3/ICE pathways, ACs were above 660.00 USD_{2019}/tCO_2eq, and GHG emission reduction levels were above 75.00% relative to the reference case. The EtOH/SOFC route presented similar AC to the NH_3/ICE route, but with a lower potential for reducing GHG emissions, while the NH_3/SOFC route presented very high levels for abatement costs, with values around 1000.00 USD_{2019}/tCO_2eq. As expected, dLUC emissions significantly changed the AC indicators for the 1G biofuel pathways. The lowest AC levels were for P-HVO/ICE (148.91 USD_{2019}/tCO_2eq), in scenarios with palm crop expansion over annual crop and pasture areas, resulting in a potential reduction of GHG emissions above 100.00%. On the other hand, the expansion of sugarcane crops over perennial crops, and of soybeans over pasture areas, reduced GHG mitigation potential to about 30.00% relative to the reference case. In these cases, the ACs showed very high levels, above 950.00 USD_{2019}/tCO_2eq. The scenarios highlight the importance of dLUC emissions to determine not only emission reduction potential but also the abatement costs of each biofuel.

The AC uncertainty results for each pathway without dLUC scenarios are presented in Fig. 9. In this case, the simulation considered the uncertainties of both lifecycle emissions and economic analyses. The results were compared with the carbon costs ranges indicated by the IPCC that can be used as an approximation of the mitigation costs in 2030, which can prevent average global temperatures from exceeding 2 °C by 2100 compared to pre-industrial levels (IPCC, 2023). In other words, the carbon costs indicate the level of economic effort needed to implement GHG mitigation policies over a certain period.

The results show three levels of variations for ACs. The lowest levels were obtained for HVO and HDPO in ICEs. An intermediate level was obtained for NH_3 in ICEs. Finally, higher levels of abatement costs were obtained for biofuels in SOFCs. In the latter case, there were high uncertainties, mainly due to uncertainties in the SOFC acquisition and replacement costs.

It was identified that the AC variation ranges were above the range of carbon cost values indicated by the IPCC. In general, the results suggest that biofuels have the potential to be an environmentally positive strategy for energy transitions in international shipping, but there are still significant challenges to overcome. In face of the climate urgency and economic difficulties, the adoption of offsetting schemes (like CORSIA) may be justified in the short-term as a complementary tool for the decarbonization of hard-to-abate sectors, like the international shipping sector. Carbon offset schemes allow individuals and companies to purchase certificates of reduction or removal (carbon credits) generated from projects developed by third parties based on pre-approved methodologies. For effective emissions mitigation, attention is focused specifically on projects and strategies that enable permanent and long-lasting removal of CO2 from the atmosphere. However, strong governmental programs and incentives will be crucial in allowing marine biofuels experience further learning effects and gains of scale, so they can bridge the competitivity gap with fossil fuels and deliver, at large scale, the positive externalities which are not autonomously recognized by the market system.

4. Conclusions

Faced with the urgency and challenges associated with the international shipping decarbonization, this work has shown the potential of biofuels potentially produced in Brazil to reduce the life-cycle carbon footprint of ships, with reductions ranging from 63.12% to 89.98% compared to the LSFO/ICE pathway. It was also highlighted the importance of dLUC scenarios to evaluate the final carbon footprint for 1G alternatives, including carbon sequestration opportunities. For SOFC technology, no benefits in terms of GHG emissions were observed among biofuels pathways, while showed important reductions in harmful emissions (PM, SOx and NOx) during ship operation. Biofuels also performed better in terms of fossil resource scarcity and fine particulate matter formation, while results for the other environmental categories varied according to the production pathway and should be considered for a comprehensive sustainability analysis. The economic indicators for biofuels pathways, however, point to a different direction, with higher MFSP and LCOS results compared to the reference case. MFSP results ranged from 2.46 times to 3.75 times the market price for LSFO and LCOS ranged from 1.58 times to 3.08 times the LCOS for LSFO/ICE pathway. Drop-in biofuels, such as HVO and HDPO, presented the lowest MFSP and LCOS values among the biofuels and, consequently, the lowest ACs levels, reinforcing their economic advantage compared to the other biofuels. Ammonia, on the other hand, showed the highest ACs levels, especially due to the high MFSP and due to the ship's CapEx increments, especially when combined with SOFC technology. This

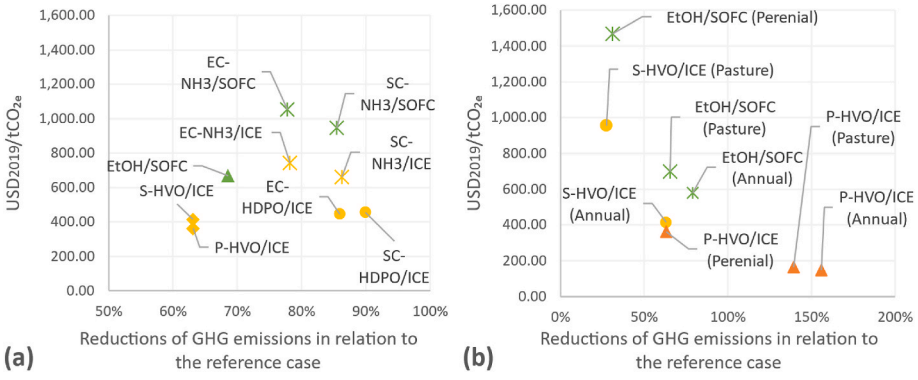


Fig. 8. Abatement costs in relation to the reference case (LSFO/ICE), (a) without dLUC, (b) considering different dLUC scenarios.

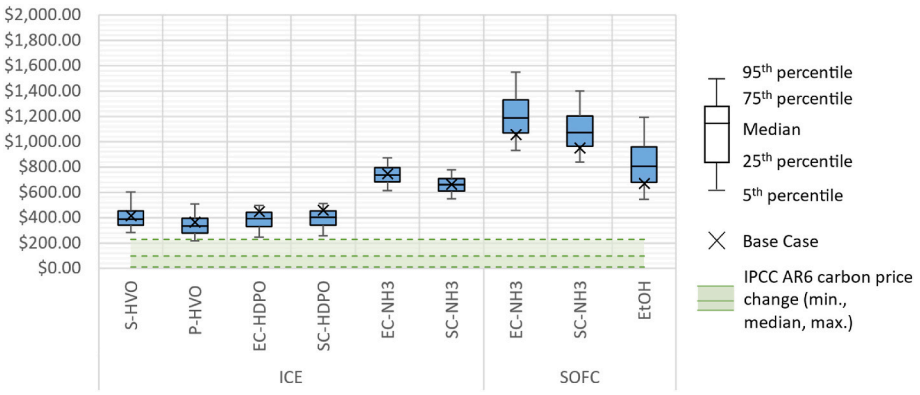


Fig. 9. Monte Carlo results for the abatement costs of each pathway relative to the reference case (LSFO/ICE) and comparisons with the variations in carbon prices from the IPCC (IPCC, 2023).

study considered some projected production conditions and the LCI were based on literature information. In this sense, some parameters used in the datasets can change over time. Monte Carlo analysis sought to highlight part of these uncertainties, and identified the main points for improvement and attention in the technical-economic and LCA modeling, that can be explored in future works, such as changes in the biofuels supply change (agricultural and industrial inputs, land management, yields, logistics), in the biorefineries capital costs and in the SOFC acquisition and replacement costs on ships. It is worth noting that this study has limitations related to the use of secondary databases in LCA modeling, which do not necessarily reflect the production conditions of the geographic and temporal scope defined in the work. Furthermore, emissions from indirect land use change have not been evaluated, as this work presents an attributional approach, and may be considered in future work, with a consequential approach. Finally, it is worth highlighting the limitations of economic modeling, which used assumptions such as interest rates, exchange rates, inflation and costs that can have wide variation in values over time and must be updated in future work.

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CRediT authorship contribution statement

Lucas Cesilla de Souza: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Joaquim Eugênio Abel Seabra:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix. Supporting Information

Details of the methodology, life-cycle datasets, technical-economic methodology, uncertainty analysis, and additional results (PDF). Supporting Information to this article can be found online at https://doi.org/10.1016/j.cesys.2024.100195.

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